Lemma A.11. *Suppose Assumption 1 holds and the σ in Assumption 1 satisfies $\sigma < \sqrt{\frac{2}{d\cdot\|(\Gamma(M_{\text{train}})\Lambda+\epsilon^2\psi(M_{\text{train}})I_d)\Lambda^{-1}\|_2}}$. Then, for any continuous training time $t \geq 0$, we have $(w_{22}^V(t))^2 \geq \nu > 0$, where*

$$\nu := \frac{\sigma^2 \cdot \|\Lambda\Theta\|_F^2 \cdot (2 - d\cdot\sigma^2 \cdot \|(\Gamma(M_{\text{train}})\Lambda + \epsilon^2\psi(M_{\text{train}})I_d)\Lambda^{-1}\|_2)}{2d\|\Lambda^2\|_2} > 0.$$

Proof. By applying Eq. (A.25) in Lemma A.10, we have that for any $t \geq 0$,

$$\begin{aligned}
&2\sigma^2 \cdot \|\Lambda\Theta\|_F^2 \cdot (d\cdot\sigma^2 \cdot \|(\Gamma(M_{\text{train}})\Lambda + \epsilon^2\psi(M_{\text{train}})I_d)\Lambda^{-1}\|_2 - 2) + 2\text{Tr}(\Lambda)\\
&\geq \tilde{\mathcal{L}}^{\text{adv}}(\theta_0) \geq \tilde{\mathcal{L}}^{\text{adv}}(\theta_t)\\
&= 2\text{Tr}\Big[(\Gamma(M_{\text{train}})\Lambda + \epsilon^2\psi(M_{\text{train}})I_d)\cdot(w_{22}^V W_{11}^{KQ}\Lambda^{\frac{1}{2}})\cdot(w_{22}^V W_{11}^{KQ}\Lambda^{\frac{1}{2}})^\top\Big]\\
&\quad - 4\text{Tr}\Big[(w_{22}^V W_{11}^{KQ}\Lambda^{\frac{1}{2}})\cdot\Lambda^{\frac{3}{2}}\Big] + 2\text{Tr}(\Lambda)\\
&= 2\|(\Gamma(M_{\text{train}})\Lambda + \epsilon^2\psi(M_{\text{train}})I_d)^{\frac{1}{2}}\cdot(w_{22}^V W_{11}^{KQ}\Lambda^{\frac{1}{2}})\|_F^2 - 4\text{Tr}\Big[w_{22}^V W_{11}^{KQ}\Lambda^2\Big] + 2\text{Tr}(\Lambda)\\
&\geq 0 - \underbrace{4d\cdot|w_{22}^V|\cdot\|\Lambda^2\|_2\cdot\|W_{11}^{KQ}\|_2}_{\text{by Lemma A.5}} + 2\text{Tr}(\Lambda),
\end{aligned}$$

which indicates

$$2\sigma^2 \cdot \|\Lambda\Theta\|_F^2 \cdot (d\cdot\sigma^2 \cdot \|(\Gamma(M_{\text{train}})\Lambda + \epsilon^2\psi(M_{\text{train}})I_d)\Lambda^{-1}\|_2 - 2) \geq -4d\cdot|w_{22}^V|\cdot\|\Lambda^2\|_2\cdot\|W_{11}^{KQ}\|_F,$$

thus

$$|w_{22}^V|\cdot\|W_{11}^{KQ}\|_F \geq \frac{\sigma^2 \cdot \|\Lambda\Theta\|_F^2 \cdot (2 - d\cdot\sigma^2 \cdot \|(\Gamma(M_{\text{train}})\Lambda + \epsilon^2\psi(M_{\text{train}})I_d)\Lambda^{-1}\|_2)}{2d\|\Lambda^2\|_2}. \quad \text{(A.26)}$$

Besides, by combining Lemma A.9 and Lemma A.10, we know that

$$w_{22}^V(t) = \sqrt{\text{Tr}[W_{11}^{KQ}(t)(W_{11}^{KQ}(t))^\top]} = \sqrt{\|W_{11}^{KQ}(t)\|_F^2} = \|W_{11}^{KQ}(t)\|_F. \quad \text{(A.27)}$$

Finally, inserting Eq. (A.27) into Eq. (A.26), we thus have

$$(w_{22}^V(t))^2 \geq \frac{\sigma^2 \cdot \|\Lambda\Theta\|_F^2 \cdot (2 - d\cdot\sigma^2 \cdot \|(\Gamma(M_{\text{train}})\Lambda + \epsilon^2\psi(M_{\text{train}})I_d)\Lambda^{-1}\|_2)}{2d\|\Lambda^2\|_2} > 0.$$

The proof is completed. □

Lemma A.12 (PL-inequality)**.** *Suppose Assumption 1 holds and the LSA model $f_{\text{LSA},\theta}$ is trained via minimizing the surrogate AT loss $\tilde{\mathcal{L}}^{\text{adv}}(\theta)$ in Eq. (9) with continuous training flow. Suppose the σ in Assumption 1 satisfies $\sigma < \sqrt{\frac{2}{d\cdot\|(\Gamma(M_{\text{train}})\Lambda+\epsilon^2\psi(M_{\text{train}})I_d)\Lambda^{-1}\|_2}}$. Then for any continuous training time $t > 0$, we uniformly have that*

$$\|\partial_\theta\tilde{\mathcal{L}}^{\text{adv}}(\theta_t)\|_2^2 \geq \mu\cdot\Big(\tilde{\mathcal{L}}^{adv}(\theta_t) - \min_\theta \tilde{\mathcal{L}}^{adv}(\theta)\Big),$$

where

$$\mu := \frac{8\nu}{\|(\Gamma_{\text{train}}\Lambda + \epsilon^2\psi_{\text{train}}I_d)^{-\frac{1}{2}}\|_F^2\cdot\|\Lambda^{-\frac{1}{2}}\|_F^2},$$

ν is defined in Lemma A.11, and $\text{Vec}(\cdot)$ denotes the vectorization function.

Proof. From Eq. (A.22) in Lemma A.9, we have that

$$\begin{aligned}
&\partial_t W_{11}^{KQ}(t) = -4\cdot(w_{22}^V)^2\cdot(\Gamma(M_{\text{train}})\Lambda + \epsilon^2\psi(M_{\text{train}})I_d)\cdot W_{11}^{KQ}\cdot\Lambda + 4w_{22}^V\Lambda^2\\
&= -4w_{22}^V\cdot(\Gamma(M_{\text{train}})\Lambda + \epsilon^2\psi(M_{\text{train}})I_d)\cdot D(\theta_t)\cdot\Lambda^{\frac{1}{2}},
\end{aligned}$$

where

$$D(\theta_t) := \Big(w_{22}^V W_{11}^{KQ} \Lambda^{\frac{1}{2}} - (\Gamma(M_{\text{train}})\Lambda + \epsilon^2 \psi(M_{\text{train}}) I_d)^{-1} \Lambda^{\frac{3}{2}}\Big) \in \mathbb{R}^{d \times d}. \tag{A.28}$$

As a result, the gradient norm square $\|\partial_\theta \tilde{\mathcal{L}}^{\text{adv}}(\theta_t)\|_2^2$ can be further lower-bounded as follows,

$$\begin{aligned}
&\|\partial_\theta \tilde{\mathcal{L}}^{\text{adv}}(\theta_t)\|_2^2 := (\partial_{w_{22}^V} \tilde{\mathcal{L}}^{\text{adv}}(\theta_t))^2 + \|\partial_{W_{11}^{KQ}} \tilde{\mathcal{L}}^{\text{adv}}(\theta_t)\|_F^2 \\
&\geq \|\partial_{W_{11}^{KQ}} \tilde{\mathcal{L}}^{\text{adv}}(\theta_t)\|_F^2 \\
&= \|4 \cdot w_{22}^V \cdot (\Gamma(M_{\text{train}})\Lambda + \epsilon^2 \psi(M_{\text{train}}) I_d) \cdot D(\theta_t) \cdot \Lambda^{\frac{1}{2}}\|_F^2 \\
&= 16 \cdot (w_{22}^V)^2 \cdot \|(\Gamma(M_{\text{train}})\Lambda + \epsilon^2 \psi(M_{\text{train}}) I_d) \cdot D(\theta_t) \cdot \Lambda^{\frac{1}{2}}\|_F^2 \\
&\geq \underbrace{16 \cdot \nu}_{\text{by Lemma A.11}} \cdot \|(\Gamma(M_{\text{train}})\Lambda + \epsilon^2 \psi(M_{\text{train}}) I_d) \cdot D(\theta_t) \cdot \Lambda^{\frac{1}{2}}\|_F^2,
\end{aligned} \tag{A.29}$$

where $\nu > 0$ is defined in Lemma A.11.

Meanwhile, according to the proof of Lemma A.8, we can rewrite and upper-bound $\Big(\tilde{\mathcal{L}}^{\text{adv}}(\theta_t) - \min_\theta \tilde{\mathcal{L}}^{\text{adv}}(\theta)\Big)$ as follows,

$$\begin{aligned}
&\Big(\tilde{\mathcal{L}}^{\text{adv}}(\theta_t) - \min_\theta \tilde{\mathcal{L}}^{\text{adv}}(\theta)\Big) \\
&= 2 \cdot \text{Tr}\Big[(\Gamma_{\text{train}}\Lambda + \epsilon^2 \psi_{\text{train}} I_d) \cdot \Big(w_{22}^V W_{11}^{KQ} \Lambda^{\frac{1}{2}} - (\Gamma_{\text{train}}\Lambda + \epsilon^2 \psi_{\text{train}} I_d)^{-1} \Lambda^{\frac{3}{2}}\Big) \\
&\qquad \cdot \Big(w_{22}^V W_{11}^{KQ} \Lambda^{\frac{1}{2}} - (\Gamma_{\text{train}}\Lambda + \epsilon^2 \psi_{\text{train}} I_d)^{-1} \Lambda^{\frac{3}{2}}\Big)^\top\Big] \\
&= 2 \cdot \text{Tr}[(\Gamma_{\text{train}}\Lambda + \epsilon^2 \psi_{\text{train}} I_d) \cdot D(\theta_t) \cdot D(\theta_t)^\top] \\
&= 2 \cdot \underbrace{\text{Tr}[(\Gamma_{\text{train}}\Lambda + \epsilon^2 \psi_{\text{train}} I_d)^{\frac{1}{2}} \cdot D(\theta_t) \cdot D(\theta_t)^\top \cdot (\Gamma_{\text{train}}\Lambda + \epsilon^2 \psi_{\text{train}} I_d)^{\frac{1}{2}}]}_{\text{Lemma A.3}} \\
&= 2 \cdot \|(\Gamma_{\text{train}}\Lambda + \epsilon^2 \psi_{\text{train}} I_d)^{\frac{1}{2}} \cdot D(\theta_t)\|_F^2 \\
&\leq 2 \cdot \|(\Gamma_{\text{train}}\Lambda + \epsilon^2 \psi_{\text{train}} I_d)^{-\frac{1}{2}}\|_F^2 \cdot \|\Lambda^{-\frac{1}{2}}\|_F^2 \cdot \|(\Gamma_{\text{train}}\Lambda + \epsilon^2 \psi_{\text{train}} I_d) \cdot D(\theta_t) \cdot \Lambda^{\frac{1}{2}}\|_F^2,
\end{aligned} \tag{A.30}$$

where $\Gamma_{\text{train}} := \Gamma(M_{\text{train}})$ and $\psi_{\text{train}} := \psi(M_{\text{train}})$.

Combining Eqs. (A.29) and (A.30), we thus know that

$$\|\partial_\theta \tilde{\mathcal{L}}^{\text{adv}}(\theta_t)\|_2^2 \geq \frac{8\nu}{\|(\Gamma_{\text{train}}\Lambda + \epsilon^2 \psi_{\text{train}} I_d)^{-\frac{1}{2}}\|_F^2 \cdot \|\Lambda^{-\frac{1}{2}}\|_F^2} \cdot \Big(\tilde{\mathcal{L}}^{\text{adv}}(\theta_t) - \min_\theta \tilde{\mathcal{L}}^{\text{adv}}(\theta)\Big).$$

The proof is completed. □

Finally, we prove Theorem 1 based on Lemma A.8 and Lemma A.12.

Proof of Theorem 1. When all the conditions hold, when the surrogate AT problem defined in Eq. (9) is solved via continuous gradient flow, by Lemma A.8 we have

$$\begin{aligned}
&\partial_t \Big(\tilde{\mathcal{L}}^{\text{adv}}(\theta_t) - \min_\theta \tilde{\mathcal{L}}^{\text{adv}}(\theta)\Big) = \partial_\theta \tilde{\mathcal{L}}^{\text{adv}}(\theta_t) \cdot \partial_t \theta_t = \partial_\theta \tilde{\mathcal{L}}^{\text{adv}}(\theta_t) \cdot (-\partial_\theta^\top \tilde{\mathcal{L}}^{\text{adv}}(\theta_t)) = -\|\partial_\theta \tilde{\mathcal{L}}^{\text{adv}}(\theta_t)\|_2^2 \\
&\leq -\mu \cdot \Big(\tilde{\mathcal{L}}^{\text{adv}}(\theta_t) - \min_\theta \tilde{\mathcal{L}}^{\text{adv}}(\theta)\Big),
\end{aligned}$$

which means

$$\Big(\tilde{\mathcal{L}}^{\text{adv}}(\theta_t) - \min_\theta \tilde{\mathcal{L}}^{\text{adv}}(\theta)\Big) \leq \Big(\tilde{\mathcal{L}}^{\text{adv}}(\theta_0) - \min_\theta \tilde{\mathcal{L}}^{\text{adv}}(\theta)\Big) \cdot e^{-\mu t}.$$

As a result, when performing continuous gradient flow optimization for an infinitely long time, since $\mu > 0$, the surrogate AT loss will eventually converge to the global minima, *i.e.*,

$$\Big(\tilde{\mathcal{L}}^{\text{adv}}(\theta_*) - \min_\theta \tilde{\mathcal{L}}^{\text{adv}}(\theta)\Big) = \lim_{t \to \infty} \Big(\tilde{\mathcal{L}}^{\text{adv}}(\theta_t) - \min_\theta \tilde{\mathcal{L}}^{\text{adv}}(\theta)\Big) \leq \Big(\tilde{\mathcal{L}}^{\text{adv}}(\theta_0) - \min_\theta \tilde{\mathcal{L}}^{\text{adv}}(\theta)\Big) \cdot \lim_{t \to \infty} e^{-\mu t} = 0,$$

where $\theta_* := \lim_{t \to \infty} \theta_t$ is the converged model parameter. Meanwhile, from Lemma A.8, we know that θ_* is a global minimizer if and only if $w_{*,22}^V W_{*,11}^{KQ} = (\Gamma(M_{\text{train}})\Lambda + \epsilon^2 \psi(M_{\text{train}}) I_d)^{-1} \Lambda$, which completes the proof. □

A.4 Proofs in Section 4.4

This section collects all proofs that omitted from Section 4.4.

Proof of Theorem 2. By substituting all M_{train} with M_{test} in proofs of Proposition 1 and Lemma A.7, we immediately have that for any model parameter θ of the LSA model $f_{\text{LSA},\theta}$,

$$\begin{aligned}\mathcal{R}(\theta, M_{\text{test}}) \leq\ & 2\text{Tr}\Big[(\Gamma(M_{\text{test}})\Lambda + \epsilon^2\psi(M_{\text{test}})I_d)\cdot(w_{22}^V W_{11}^{KQ}\Lambda^{\frac{1}{2}})\cdot(w_{22}^V W_{11}^{KQ}\Lambda^{\frac{1}{2}})^\top\Big] \\ & - 4\text{Tr}\Big[(w_{22}^V W_{11}^{KQ}\Lambda^{\frac{1}{2}})\cdot\Lambda^{\frac{3}{2}}\Big] + 2\text{Tr}(\Lambda).\end{aligned}$$

By inserting the converged model parameter $\theta_*(M_{\text{train}})$, which satisfies $(w_{*,22}^V W_{*,11}^{KQ}) = (\Gamma(M_{\text{train}})\Lambda + \epsilon^2\psi(M_{\text{train}})I_d)^{-1}\Lambda$, into the above robust generalization bound, we thus have that

$$\begin{aligned}&\mathcal{R}(\theta_*(M_{\text{train}}), M_{\text{test}}) \\ &\leq 2\text{Tr}\Big[(\Gamma(M_{\text{test}})\Lambda + \epsilon^2\psi(M_{\text{test}})I_d)\cdot((\Gamma(M_{\text{train}})\Lambda + \epsilon^2\psi(M_{\text{train}})I_d)^{-1}\Lambda\cdot\Lambda^{\frac{1}{2}}) \\ &\qquad\cdot((\Gamma(M_{\text{train}})\Lambda + \epsilon^2\psi(M_{\text{train}})I_d)^{-1}\Lambda\cdot\Lambda^{\frac{1}{2}})^\top\Big] \\ &\quad - 4\text{Tr}\Big[(\Gamma(M_{\text{train}})\Lambda + \epsilon^2\psi(M_{\text{train}})I_d)^{-1}\Lambda\cdot\Lambda^{\frac{1}{2}}\cdot\Lambda^{\frac{3}{2}}\Big] + 2\text{Tr}(\Lambda) \\ &\overset{(*)}{\leq} 2\text{Tr}\Big[(\Gamma(M_{\text{test}})\Lambda + \epsilon^2\psi(M_{\text{test}})I_d)\cdot(\Gamma(M_{\text{train}})\Lambda + \epsilon^2\psi(M_{\text{train}})I_d)^{-1} \\ &\qquad\cdot\Lambda^3\cdot((\Gamma(M_{\text{train}})\Lambda + \epsilon^2\psi(M_{\text{train}})I_d)^{-1})^\top\Big] + 0 + 2\text{Tr}(\Lambda) \\ &\overset{(**)}{\leq} 2\text{Tr}\Big[\Lambda^3\cdot(\Gamma(M_{\text{test}})\Lambda + \epsilon^2\psi(M_{\text{test}})I_d)\cdot(\Gamma(M_{\text{train}})\Lambda + \epsilon^2\psi(M_{\text{train}})I_d)^{-2}\Big] + 2\text{Tr}(\Lambda),\end{aligned}$$

where $(*)$ is due to that the matrix $((\Gamma(M_{\text{train}})\Lambda + \epsilon^2\psi(M_{\text{train}})I_d)^{-1}\Lambda^3)$ is positive definite, and $(**)$ is due to that: (1) $(\Gamma(M_{\text{train}})\Lambda + \epsilon^2\psi(M_{\text{train}})I_d)^{-1}$ is symmetric and is commutative with Λ^3, and (2) Lemma A.3.

The proof is completed. □

Proof of Corollary 1. Let $\lambda_1, \cdots, \lambda_d$ be the d singular values of the matrix Λ. Then, the robust generalization bound in Theorem 2 can be rewritten as follows,

$$\begin{aligned}&\mathcal{R}(\theta_*(M_{\text{train}}), M_{\text{test}}) \\ &\leq 2\text{Tr}\Big[\Lambda^3\cdot(\Gamma(M_{\text{test}})\Lambda + \epsilon^2\psi(M_{\text{test}})I_d)\cdot(\Gamma(M_{\text{train}})\Lambda + \epsilon^2\psi(M_{\text{train}})I_d)^{-2}\Big] + 2\text{Tr}(\Lambda), \\ &\leq \sum_{i=1}^d \lambda_i^3\cdot\frac{\frac{N+M_{\text{test}}+1}{N+M_{\text{test}}}\lambda_i + \frac{\text{Tr}(\Lambda)}{N+M_{\text{test}}} + \epsilon^2\cdot\frac{M_{\text{test}}^2\text{Tr}(\Lambda)}{(N+M_{\text{test}})^2}}{\left(\frac{N+M_{\text{train}}+1}{N+M_{\text{train}}}\lambda_i + \frac{\text{Tr}(\Lambda)}{N+M_{\text{train}}} + \epsilon^2\cdot\frac{M_{\text{train}}^2\text{Tr}(\Lambda)}{(N+M_{\text{train}})^2}\right)^2} + 2\text{Tr}(\Lambda) \\ &\leq \sum_{i=1}^d \lambda_i^3\cdot\frac{\frac{N+M_{\text{test}}+1}{N+M_{\text{test}}}\lambda_i + \frac{\text{Tr}(\Lambda)}{N+M_{\text{test}}}}{\left(\frac{N+M_{\text{train}}+1}{N+M_{\text{train}}}\lambda_i\right)^2} + \sum_{i=1}^d \lambda_i^3\cdot\frac{\epsilon^2\cdot\frac{M_{\text{test}}^2\text{Tr}(\Lambda)}{(N+M_{\text{test}})^2}}{\left(\epsilon^2\cdot\frac{M_{\text{train}}^2\text{Tr}(\Lambda)}{(N+M_{\text{train}})^2}\right)^2} + 2\text{Tr}(\Lambda) \\ &\leq \sum_{i=1}^d \lambda_i\cdot\left(\frac{N+M_{\text{train}}}{N+M_{\text{train}}+1}\right)^2\cdot\left(\frac{N+M_{\text{test}}+1}{N+M_{\text{test}}}\lambda_i + \frac{\sum_{k=1}^d\lambda_k}{N}\right) \\ &\quad + \sum_{i=1}^d \frac{\lambda_i^3}{\epsilon^2\cdot\max_{k=1}^d\{\lambda_k\}}\cdot\frac{(N+M_{\text{train}})^4}{N^2}\cdot\frac{M_{\text{test}}^2}{M_{\text{train}}^4} + 2\sum_{i=1}^d\lambda_i \\ &\leq \mathcal{O}(d)\cdot\mathcal{O}(1)\cdot\left(\mathcal{O}(1) + \frac{\mathcal{O}(d)}{N}\right) + \mathcal{O}(d)\cdot\mathcal{O}\left(\frac{1}{\epsilon^2}\right)\cdot\frac{(N+M_{\text{train}})^4}{N^2}\cdot\frac{M_{\text{test}}^2}{M_{\text{train}}^4} + \mathcal{O}(d) \\ &\leq \mathcal{O}(d) + \mathcal{O}\left(\frac{d^2}{N}\right) + \mathcal{O}\left(\frac{d}{\epsilon^2}\right)\cdot\frac{(N+M_{\text{train}})^4}{N^2}\cdot\frac{M_{\text{test}}^2}{M_{\text{train}}^4}.\end{aligned}$$